

Student Management System

User Story:

The Student Enrollment & Course Management System is a comprehensive Salesforce application designed to streamline and automate the academic enrollment operations of an educational institute. This system enables administrators, enrollment officers, and instructors to efficiently manage student records, course offerings, enrollments, instructor assignments, approval processes, payments, and reporting—all within a unified Salesforce environment.

1. INTRODUCTION

1.1 Project Overview :

This project aims to centralize all student, course, and enrollment-related data within a single Salesforce application to ensure easy data access, structured workflows, and accurate reporting. The system allows Enrollment Officers to manage student admissions, course registrations, and communication effectively, while instructors and administrators gain full transparency into course allocations and academic progress. The solution also enables automated instructor assignment, task creation, validation checks, and multi-level approval processes for enrollment management.

With Salesforce Flows, Approval Processes, Validation Rules, Reports, and Dashboards, the organization can ensure that all enrollment operations—from admission to course completion—are handled systematically and with continuous monitoring of key performance indicators such as enrollment volume, revenue, student status distribution, and pending approvals.

1.2 Purpose :

The primary purpose of SECMS is to replace manual and fragmented enrollment

processes with a scalable, automated, and secure Salesforce solution that ensures:

Accurate student data management
Structured enrollment workflows
Automated approvals and validations
Transparent reporting and monitoring

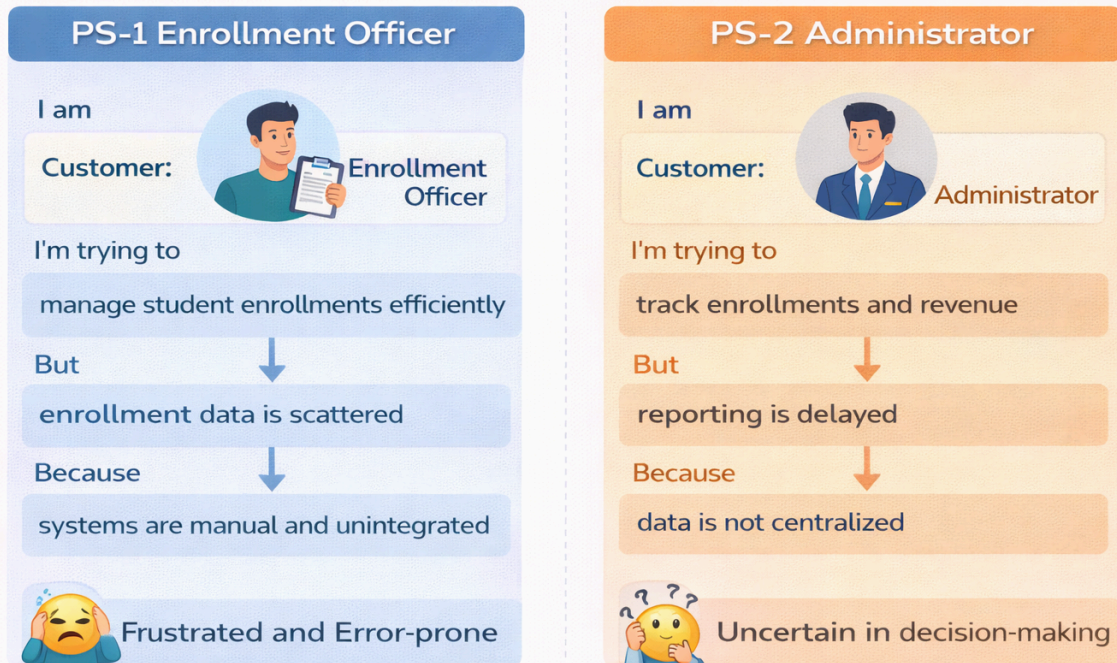
2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Enrollment Officer	manage student enrollment s efficiently	enrollment data is scattered	systems are manual and unintegrated	frustrated and error-prone
PS-2	Administrator	track enrollment s and revenue	reporting is delayed	data is not centralized	uncertain in decision-making

2. IDEATION PHASE

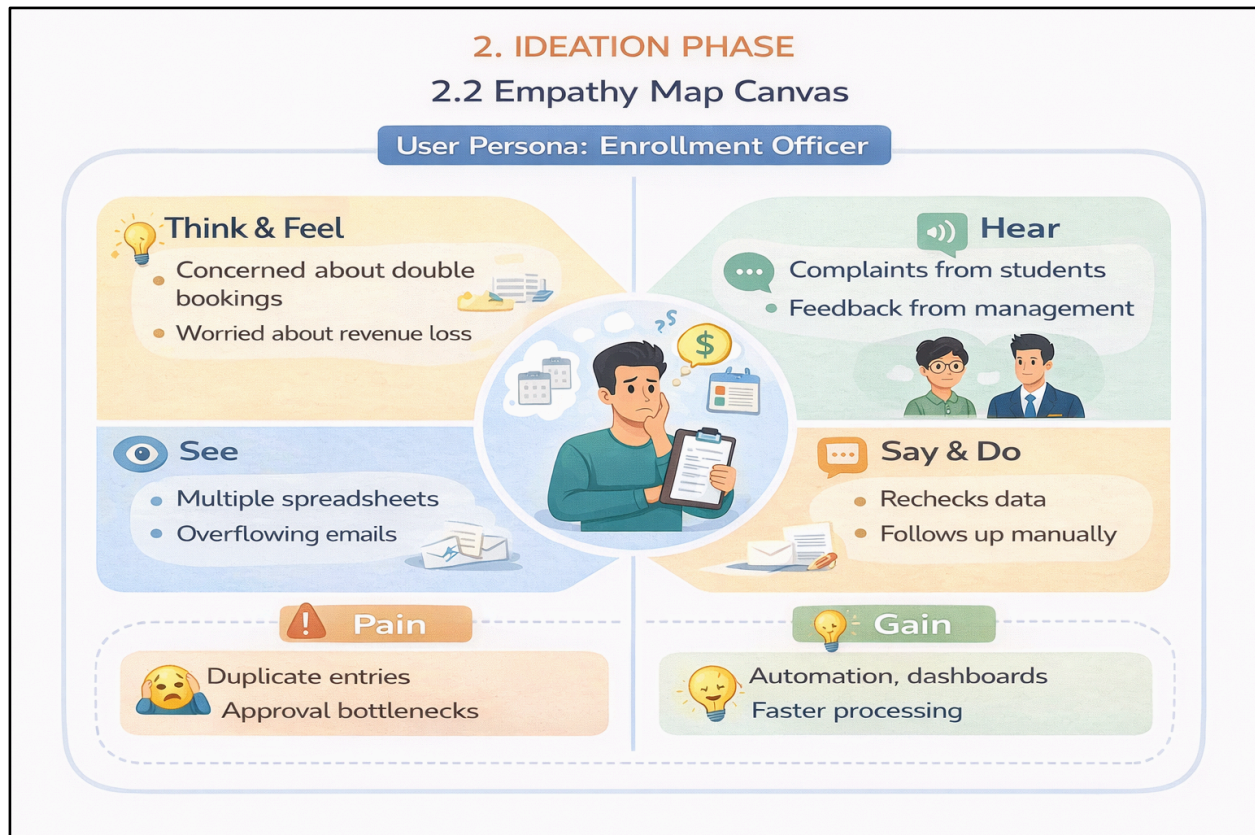
2.1 Problem Statement (Customer Problem Statement Template)



2.2 Empathy Map Canvas

User Persona: Enrollment Officer

- **Think & Feel:** Concerned about double bookings and revenue loss
- **Hear:** Complaints from students and management
- **See:** Multiple spreadsheets and emails
- **Say & Do:** Rechecks data, follows up manually
- **Pain:** Duplicate entries, approval bottlenecks
- **Gain:** Automation, dashboards, faster processing



2.3 Brainstorming & Idea Prioritization

Step 1 – Brainstormed Ideas

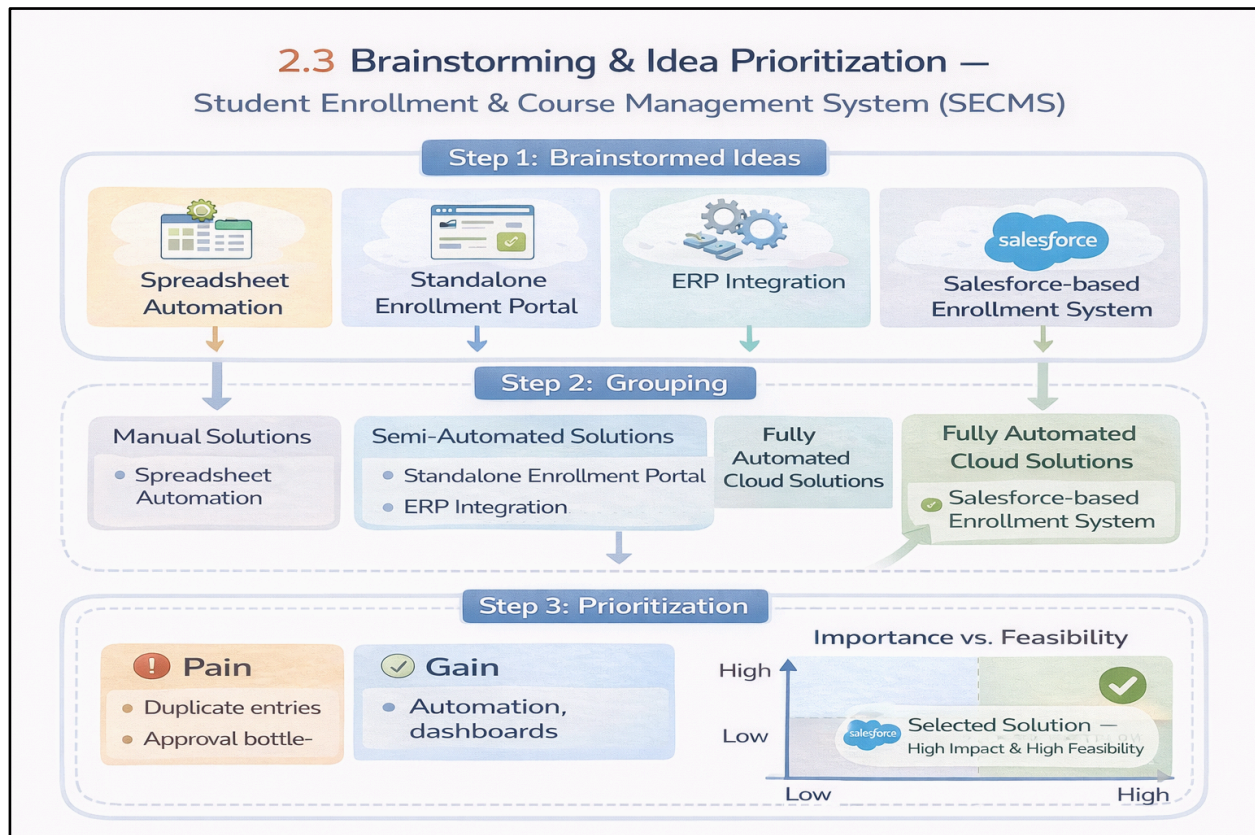
- Spreadsheet automation
- Standalone enrollment portal
- ERP integration
- Salesforce-based enrollment system

Step 2 – Grouping

- Manual solutions
- Semi-automated solutions
- Fully automated cloud solutions

Step 3 – Prioritization

Salesforce-based SECMS was selected due to **high feasibility** and **high impact**.



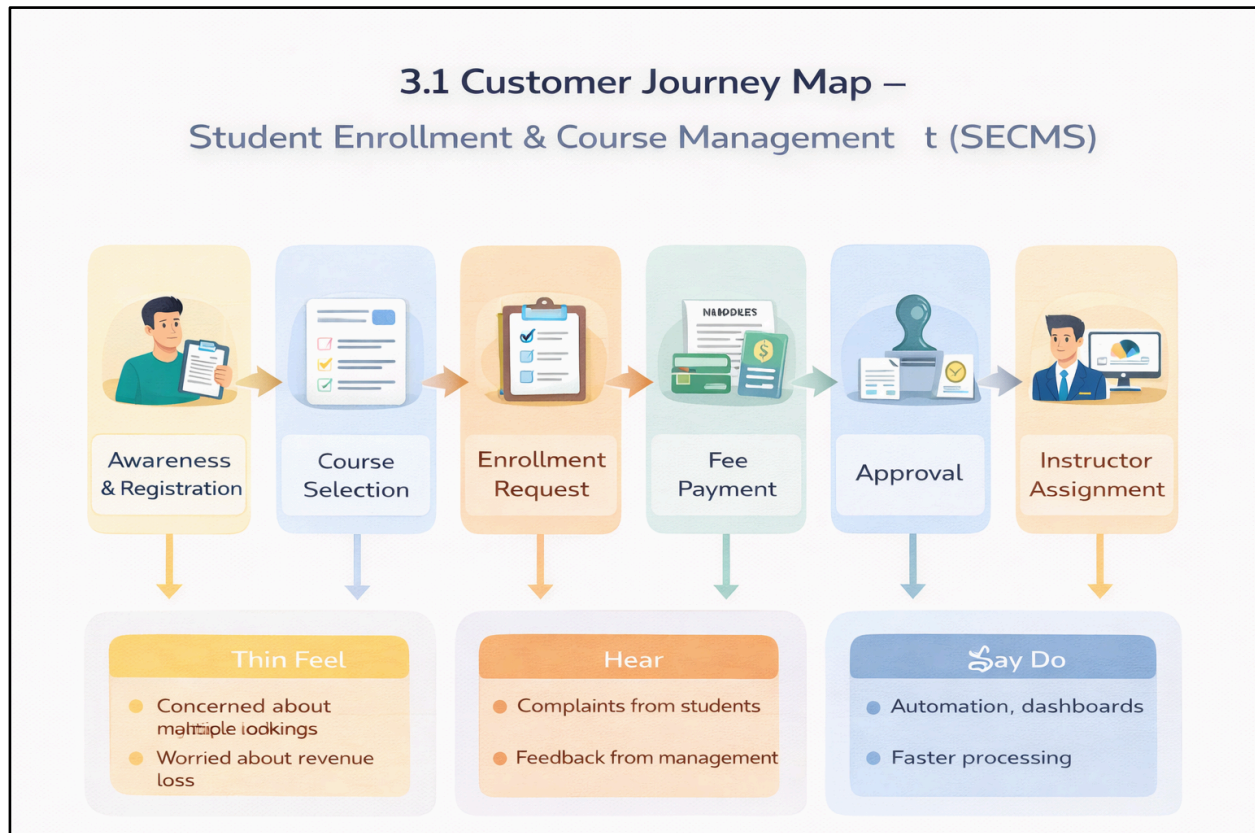
3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages covered:

- Awareness & Registration
- Course Selection
- Enrollment Request
- Fee Payment

- Approval
- Instructor Assignment
- Reporting & Completion



3.2 Solution Requirements

Functional Requirements

FR No	Epic	Description
-------	------	-------------

FR-1	Student Management	Manage student records
FR-2	Course Management	Create and manage courses
FR-3	Enrollment Processing	Request, approve, reject enrollments
FR-4	Instructor Assignment	Auto-assign instructors
FR-5	Reporting	Dashboards and reports

Non-Functional Requirements

NFR-1: Usability

The system shall provide a simple, intuitive, and user-friendly interface that enables enrollment officers, administrators, instructors, and students to perform operations efficiently with minimal training. Navigation shall be organized through clearly labeled tabs, dashboards, and forms to reduce user effort and errors.

Examples:

- Quick student registration form
- One-click enrollment approval
- Role-based dashboards

NFR-2: Security

The system shall ensure the confidentiality, integrity, and protection of student and institutional data through secure authentication, authorization, and access controls. Sensitive information such as personal data, enrollment history, and academic records shall only be accessible to authorized users.

Examples:

- Only admins can modify course details
- Instructors view only assigned students
- Students cannot access other student records

NFR-3: Reliability

The system shall operate consistently and accurately without failures during enrollment processing, approvals, and reporting. Automation logic shall execute correctly to avoid duplicate or incorrect enrollments.

Examples:

- No duplicate course registration
- Accurate status updates after approvals
- Consistent report generation

NFR-4: Performance

The system shall provide fast response times for all operations to ensure smooth and uninterrupted user experience, especially during peak admission periods.

Examples:

- Quick dashboard refresh
- Fast search for student records
- No lag during bulk uploads

NFR-5: Scalability

The system shall support future growth in the number of students, courses, departments, and users without degradation in performance or functionality. It shall allow easy addition of new modules and features.

Examples:

- Add new departments without redesign
- Handle increased admissions annually
- Extend to mobile or portal integration

NFR-6: Availability

The system shall be continuously accessible to users during institutional working hours and must minimize downtime. The platform shall support high availability for academic operations.

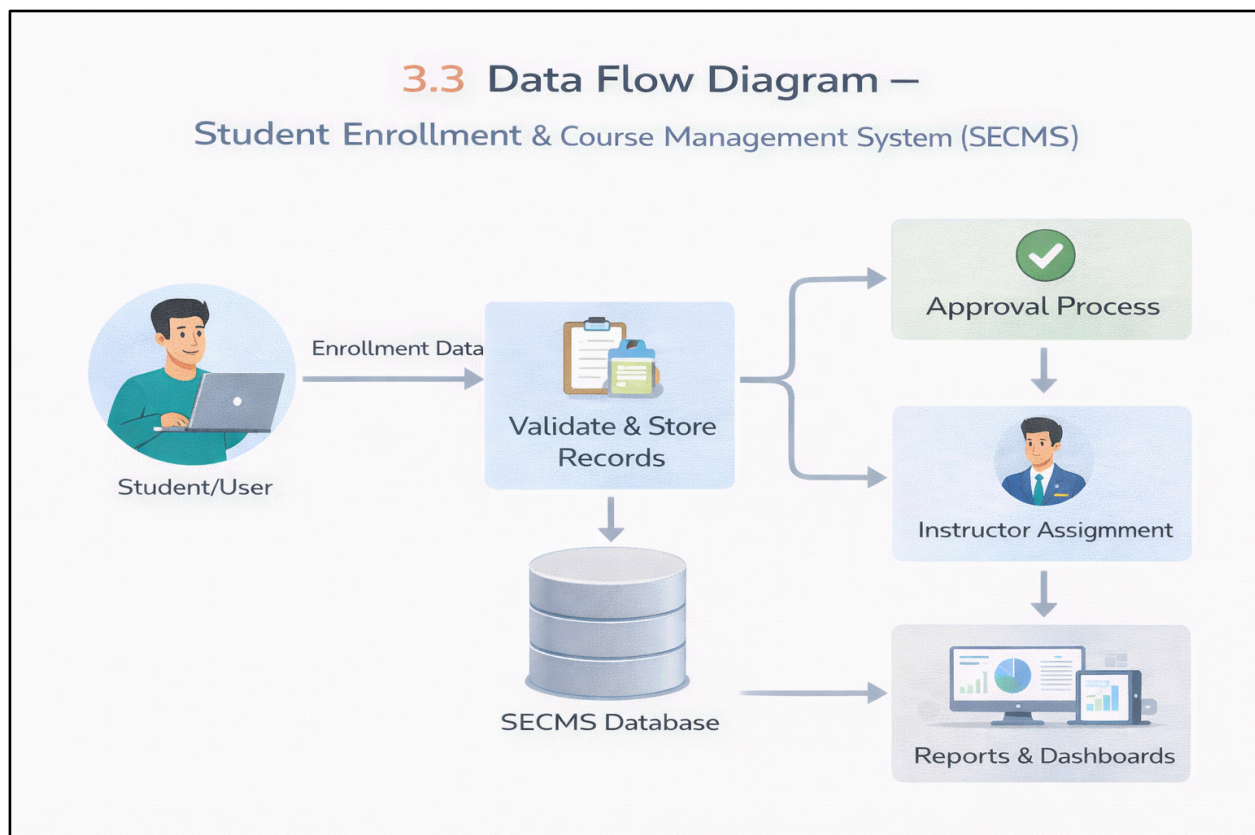
Examples:

- 24/7 cloud access
- Reliable during peak admission season
- Automatic backup of student records

3.3 Data Flow Diagram

- User inputs student and enrollment data
- System validates and stores records
- Approval process routes data
- Reports and dashboards consume processed data.

The system processes enrollment requests, validates data, routes approvals, assigns instructors, and generates reports.



3.4 Technology Stack

Layer	Technology
UI	Salesforce Lightning
Logic	Salesforce Flows
Database	Salesforce Objects
Automation	Validation Rules, Approval Process
Reporting	Reports & Dashboards

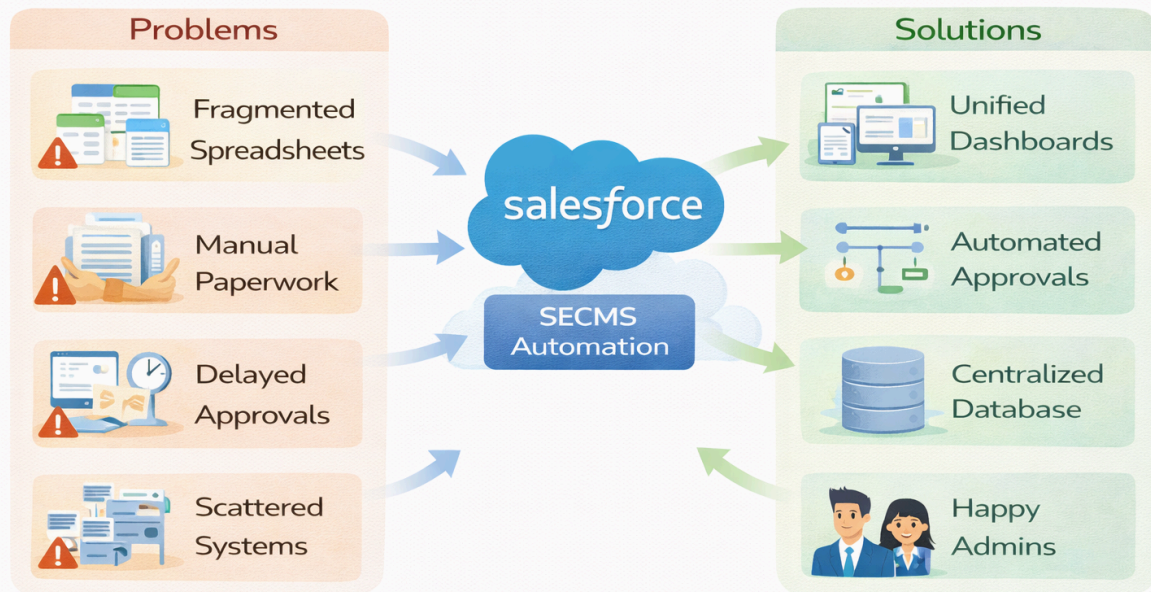
4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

SECMS directly addresses fragmented enrollment workflows by aligning user behavior, constraints, and automation using Salesforce-native capabilities.

4.1 Problem–Solution Fit

– Student Enrollment & Course Management System (SECMS)



4.2 Proposed Solution

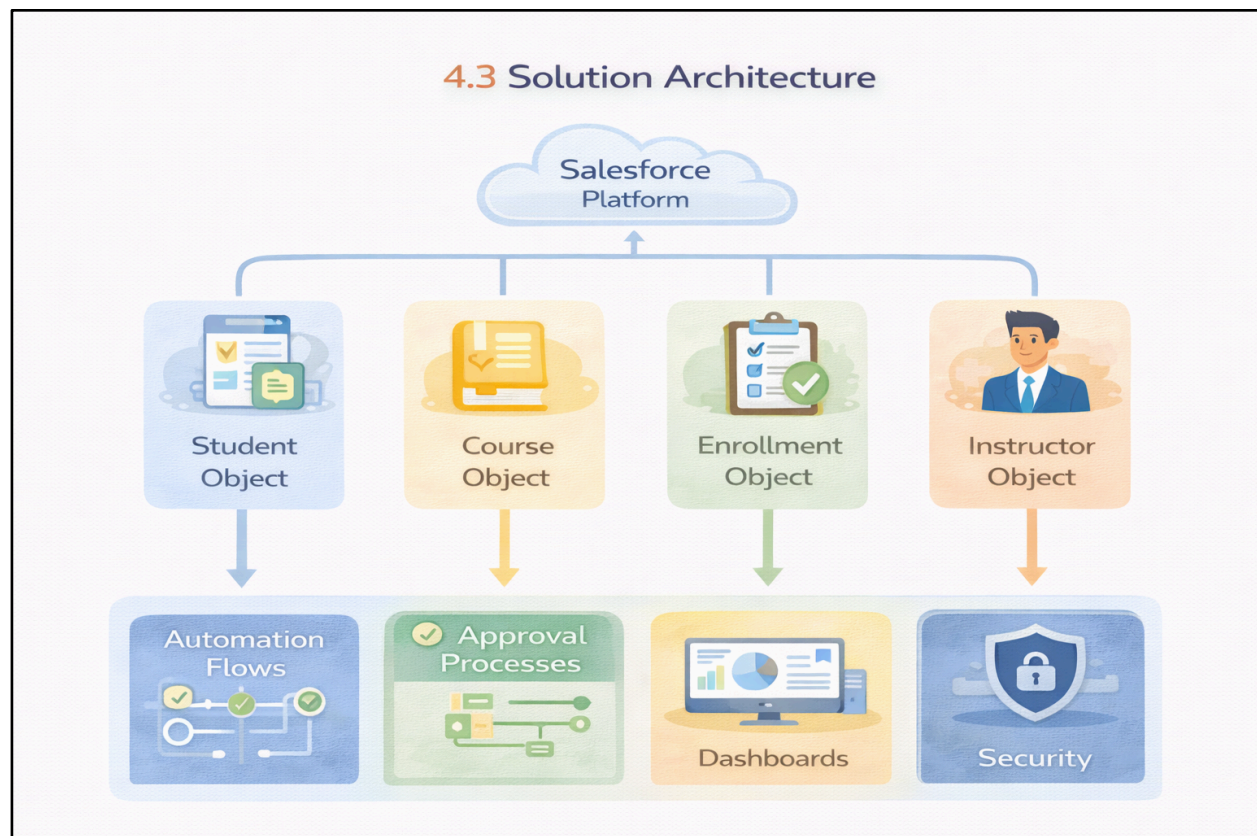
Functional Requirements

Parameter	Description
Problem	Manual enrollment handling
Solution	Salesforce-based SECMS
Novelty	End-to-end automation

Social Impact	Improved student experience
Revenue Model	Institutional deployment
Scalability	Multi-department support

4.3 Solution Architecture

The architecture integrates Salesforce objects, automation flows, approval processes, dashboards, and security layers.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Developer Setup	USN-1	As a system administrator, I want to create a Salesforce developer account so that I can configure and deploy the Student Enrollment system..	1	High	Member1
Sprint-1	Data Modeling	USN-2	As an admin, I want to create custom objects for Student, Course, Enrollment, and Instructor so that academic data can be stored and managed efficiently.	3	High	Member2
Sprint-2	Data Modeling	USN-3	As a user, I want dedicated tabs for each object so that I can easily navigate between students, courses, enrollments, and instructors.	3	High	Member3
Sprint-2	Data Modeling	USN-4	As an admin, I want to create relevant fields such as student details, course information, and enrollment status so that complete records can be captured accurately.	5	High	Member4
Sprint-2	Data Modeling	USN-5	As an enrollment officer, I want a centralized Lightning App so that I can access all academic operations from one interface.	3	High	Member4
Sprint-2	Data Modeling	USN-6	As an admin, I want to define relationships between students, courses, and enrollments so that the system can logically link and track academic activities.	5	Medium	Member1
Sprint-2	Data Modeling	USN-7	As a user, I want customized page layouts so that only relevant information is displayed, improving clarity and ease of data entry.	5	High	Member2

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-3	Automation	USN-8	As a system, I want to validate input data such as mandatory fields and enrollment limits so that incorrect or incomplete records are prevented.	3	High	Member4
Sprint-3	Automation	USN-9	As a system, I want automated flows to handle enrollment processing and notifications so that manual effort is reduced and processes become faster.	3	High	Member4
Sprint-3	Automation	USN-10	As an administrator, I want an approval process for enrollment requests so that registrations are verified before confirmation.	5	High	Member1
Sprint-4	Security	USN-11	As an admin, I want to configure role-based security so that users can only access data relevant to their responsibilities.	5	High	Member2
Sprint-5	Reports	USN-12	As a manager, I want reports on student enrollments, course registrations, and instructor assignments so that I can monitor academic operations.	4	High	Member3
Sprint-6	Dashboards	USN-13	As a manager, I want visual dashboards showing enrollment statistics and trends so that I can quickly analyze system performance.	4	Medium	Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	6 Days	24 Oct 2022	29 Oct 2022	20	29 Oct 2022
Sprint-2	20	6 Days	31 Oct 2022	05 Nov 2022		
Sprint-3	20	6 Days	07 Nov 2022	12 Nov 2022		
Sprint-4	20	6 Days	14 Nov 2022	19 Nov 2022		

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)

6. Project Development Phase:

Project Flow:

Milestone 1: Salesforce Developer Account Creation

Milestone 2: Custom Object Creation (Student, Course, Enrollment, Instructor)

Milestone 3: Tabs Creation

Milestone 4: Lightning App Creation

Milestone 5: Fields Creation

Milestone 6: Create Relationships

Milestone 7: Page Layout Customization

Milestone 8: Validation Rules

Milestone 9: Flows (Automation)

Milestone 10: Approval Process

Milestone 11: Reports

Milestone 12: Dashboards

Milestone 13: Security Setup

Milestone 14: Conclusion

What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows

7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, and sharing
9. End-to-end application testing

Milestone 1-Salesforce :

Introduction:

Are you new to Salesforce? Not sure exactly what it is, or how to use it? Don't know where you should start on your learning journey? If you've answered yes to any of these questions, then you're in the right place. This module is for you.

Welcome to Salesforce! Salesforce is game-changing technology, with a host of productivity-boosting features, that will help you sell smarter and faster. As you work toward your badge for this module, we'll take you through these features and answer the question, "What is Salesforce, anyway?".

What Is Salesforce?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud. So what does that really mean? Well, before Salesforce, your contacts, emails, follow-up tasks, and prospective deals might have been organized something like this:

<https://youtu.be/r9EX3lGde5k>

Activity 1: Creating Developer Account:

Creating a developer org in salesforce.

Go to <https://developer.salesforce.com/signup>

On the sign up form, enter the following details :

Build enterprise-quality apps fast to bring your ideas to life

- Build apps fast with drag and drop tools
- Customize your data model with clicks
- Go further with Apex code
- Integrate with anything using powerful APIs
- Stay protected with enterprise-grade security
- Customize UI with clicks or any leading-edge web framework

Sign up for your Salesforce Developer Edition
A full-featured copy of the Platform, for free

Complete the form to start your free trial. Our team will be in touch to help you make the most of your trial.

First Name* Last Name*

Email*

Role*

Company*

First name & Last name

Email

Role : Developer

Company : College Name

County : India

Postal Code : pin code

Username : should be a combination of your name and company

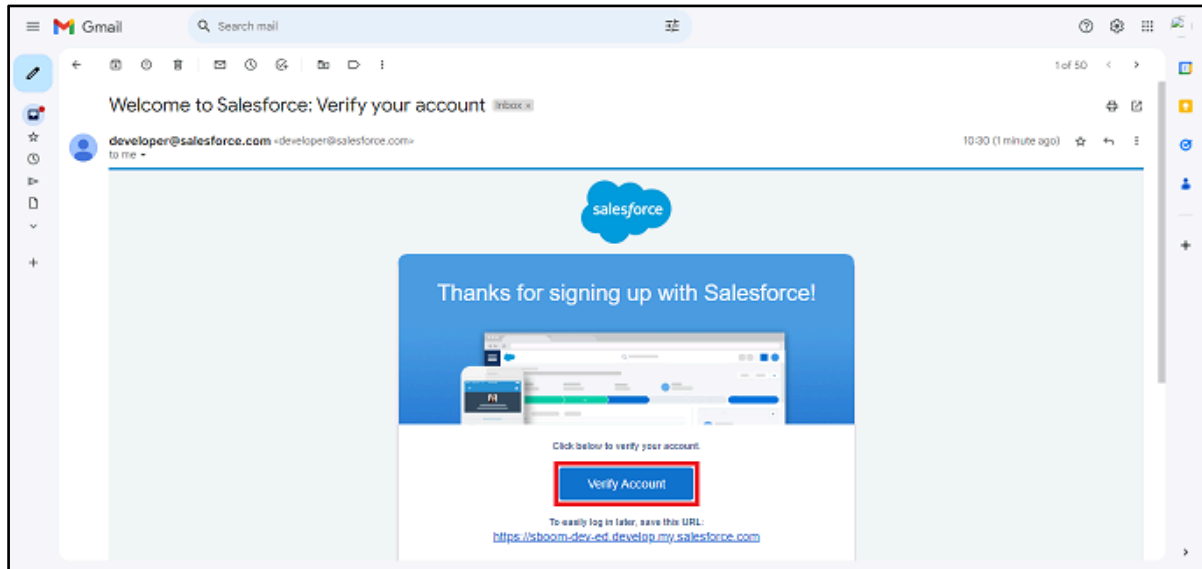
This need not be an actual email id, you can give anything in the format :

username@organization.com

Click on sign me up after filling these.

Activity 2: Account Activation:

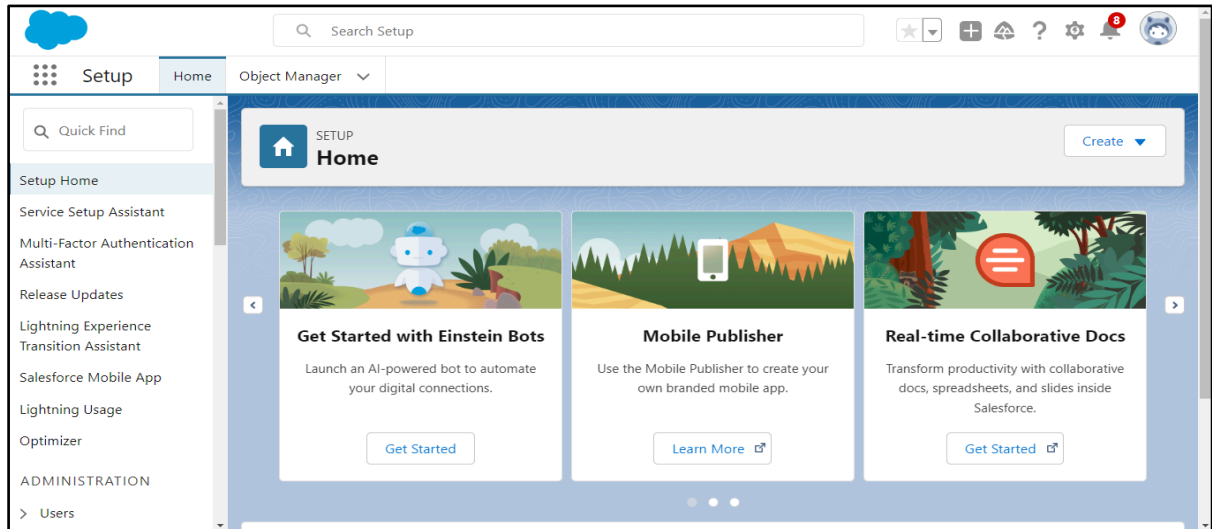
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 5-10mins.



2. Click on Verify Account
3. Give a password and answer a security question and click on change password.

A screenshot of the 'Change Your Password' page in Salesforce. The page prompts the user to enter a new password for 'lead@sb.com'. It lists requirements: 8 characters, 1 letter, and 1 number. The 'New Password' field is filled with 'Good', the 'Confirm New Password' field is filled with 'Match', and the 'Security Question' is 'In what city were you born?'. The 'Answer' field is filled with 'asdfghjkl'. The 'Change Password' button is highlighted with a red rectangle.

4. Then you will redirect to your salesforce setup page.



Milestone 2 – Objects:

Activity 1: Creating Student Object:

The purpose of creating the Student custom object is to store and manage information about students such as their personal details, contact information, and enrollment status.

To create the Student object:

Go to the Setup page.

Click on Object Manager.

Click on Create → Custom Object.

Enter the details as below:

1. Label: Student
2. Plural Label: Students
3. Object Name: Student
4. Record Name: Student Name

5. Data Type: Text
6. Under Optional Features, select:
7. Allow Reports
8. Allow Activities (optional but recommended)
9. Track Field History (optional, if you want)
10. Under Search Status, select:
11. Allow Search
12. Click Save.

Setup Home Object Manager

SETUP > OBJECT MANAGER

Student

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules

Custom Object Definition Edit Save Save & New Cancel

Custom Object Information

The singular and plural labels are used in tabs, page layouts, and reports.
Be careful when changing the name or label as it may affect existing integrations and merge templates.

Label Student Example: Account

Plural Label Students Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Student Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Content Name --None--

SetupHomeObject Manager

SETUP > OBJECT MANAGER

Student

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Enter Record Name Label and Format

The Record Name appears in page layouts, key lists, related lists, lookups, and search results. For example, the Record Name for Account is "Account Name" and for Case it is "Case Name" when referenced via the API.

Record NameStudent NameExample: Account Name

Data typeTextWarning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type.

Optional Features

☒ Allow Reports

☒ Allow Activities

☒ Track Field History

☐ Allow in Chatter Groups

☐ Enable Licensing

Object Classification

When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object.

☒ Allow Sharing

☒ Allow Bulk API Access

☒ Allow Streaming API Access

Deployment Status

☐ In Development

☒ Deployed

Deployment Status

☐ In Development

☒ Deployed

Search Status

When this setting is enabled, your users can find records of this object type when they search. [Learn more.](#)

☒ Allow Search

SaveSave & NewCancel

In the same way Create **Courses, Instructors, and Enrolments** objects

Milestone 3 - Fields:

Table 1: Student Object Fields:

Field Label	Data Type
-------------	-----------

Student Name	Text (Standard)
Email	Email
Phone	Phone
Date of Birth	Date
Student Status	Picklist
Address	Text Area

Table 2: Course Object Fields:

Field Label	Data Type
Course Name	Text (Standard)
Category	Picklist
Duration	Number/Text
Course Fee	Currency
Description	Text Area

Table 3: Instructor Object Fields:

Field Label	Data Type
Instructor Name	Text (Standard)
Instructor	Text

Code	
Expertise	Picklist
Phone	Phone
Email	Email

Table 4: Enrollment Object Fields:

Field Label	Data Type
Student	Lookup (Student)
Course	Lookup (Course)
Instructor	Lookup (Instructor)
Enrollment Date	Date
Enrollment Status	Picklist
Fees Paid	Checkbox
Total Amount	Currency
Comments	Text Area

Milestone 4 – Create Relationships:

1. Activity 1: Create Lookup from Enrollment to Student:

1.

Go to Setup → click Object Manager.

2. Type Enrollment in Quick Find → click Enrollment__c object.

3. Click Fields & Relationships → click New.

4. Select Lookup Relationship → click Next.

5. For Related To, select Student__c → click Next.

6. Enter Field Label: Student (API Name: Student__c).

7. Click Next through Field-Level Security and Layout options (ensure it's visible and added to layout).

8. Click Save.

Edit Enrollment Custom Field
Student

Step 1. Choose the field type

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected ☒ Lookup Relationship

Select one of the data types below.

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Next Cancel

Previous Save Next Cancel

Field Label Student

Field Name Student

Description

Help Text

Child Relationship Name Enrollments

Required ☒ Many-to-one relationship. Many-to-one relationships require a value in this field in order to save a record.

do if the lookup record is deleted?

☐ Clear the value of this field. You can't choose this option if you make this field required.

☒ Don't allow deletion of the lookup record that's part of a lookup relationship.

Activity 2: Create Lookup from Enrollment to Course

Purpose:

To link each Enrollment to a Course.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Select Lookup Relationship → Next.
4. Related To: select Course__c → click Next.
5. Field Label: Course (API Name: Course__c).
6. Click Next (set visibility and layout) → Save.

Edit Enrollment Custom Field
Course

Step 1. Choose the field type

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected

☒ Lookup Relationship

Select one of the data types below.

Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Next Cancel

Edit Relationship
Enrollment

Step 2. Enter the label and name for the lookup field

Field Label Course

Field Name Course

Description

Help Text

Child Relationship Name Enrollments

Required ☒ Always require a value in this field in order to save a record

What to do if the lookup record is deleted? ☐ Clear the value of this field. You can't choose this option if you make this field required.

Previous Save Next Cancel

Activity 3: Create Lookup from Enrollment to Instructor:

Purpose:

To store which Instructor is assigned to a particular Enrollment.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.
2. Click Fields & Relationships → New.
3. Choose Lookup Relationship → Next.
4. Related To: select Instructor__c → Next.
5. Field Label: Instructor (API Name: Instructor__c).
6. Complete Field-Level Security and Page Layout options → Save

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Specify the type of information that the custom field will contain.

Data Type

☐ None Selected Select one of the data types below.

☐ Auto Number A system-generated sequence number that uses a display format you define. The number is automatically incremented for each new record.

☐ Formula A read-only field that derives its value from a formula expression you define. The formula field is updated when any of the source fields change.

☐ Roll-Up Summary A read-only field that displays the sum, minimum, or maximum value of a field in a related list or the record count of all records listed in a related list.

☒ **Lookup Relationship** Creates a relationship that links this object to another object. The relationship field allows users to click on a lookup icon to select a value from a popup list. The other object is the source of the values in the list.

Next Cancel

Enrollment

New Relationship

Help for this Page

Step 2. Choose the related object Step 2

Previous Next Cancel

Select the other object to which this object is related.

Related To **Instructor**

Previous Next Cancel

Step 3. Enter the label and name for the lookup field Step 3 of 6

Previous Next Cancel

Field Label **Instructor**

Field Name **Instructor**

Description

Help Text

Child Relationship Name **Enrollments**

Required ☒ Always require a value in this field in order to save a record

What to do if the lookup record is deleted? ☐ Clear the value of this field. You can't choose this option if you make this field required.

☒ Don't allow deletion of the lookup record that's part of a lookup relationship.

Auto add to custom report type ☒ Add this field to existing custom report types that contain this entity

Enrollment
New Relationship

Help for this Page

Step 6. Add custom related lists Step 6 of 6

Previous Save & New **Save** Cancel

Field Label	Instructor
Data Type	Lookup
Field Name	Instructor
Description	

Specify the title that the related list will have in all of the layouts associated with the parent.

Related List Label

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout. To change the values that appear, you will need to customize the Record Types.

Add Related List	Page Layout Name
☒	Instructor Layout

☒ Append related list to users' existing personal customizations

Milestone 5 – Validation Rules:

Activity 1: Student Email Domain Validation

Purpose:

To ensure students use only allowed institutional email addresses.

Steps:

Go to Setup → Object Manager → Student__c.

Click Validation Rules → New.

Rule Name: Student_Email_Domain_Validation.

Error Condition Formula: (example for @student.edu)

- NOT(CONTAINS(Email__c, "@student.edu"))

Error Message: Email must be a valid student institutional email.

Error Location: Field → Email__c.

Click Save.

Setup Home Object Manager

SETUP > OBJECT MANAGER

Student

- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters
- Restriction Rules
- Scoping Rules
- Object Access
- Triggers
- Flow Triggers
- Validation Rules**

Validation Rule Edit

Save Save & New Cancel

Rule Name **Email_Must_Be_Student_Domain**

Active ☒

Description

Error Condition Formula

Example: `Discount_Percent__c > 0.30` [More Examples...](#)
 Display an error if Discount is more than 30%
 If this formula expression is **true**, display the text defined in the Error Message area

Insert Field Insert Operator

NOT(CONTAINS(Email__c, "@student.edu"))

Check Syntax

Functions

-- All Function Categories --

ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

Insert Selected Function

ABS(number)
Returns the absolute value of a number, a number without its sign

[Help on this function](#)

Error Message

Example: `Discount percent cannot exceed 30%`

This message will appear when Error Condition formula is **true**

Error Message **Email must be a valid student institution email**

This error message can either appear at the top of the page or below a specific field on the page

Error Location ☐ Top of Page ☒ Field **Email**

Save Save & New Cancel

Activity 2: Fees Must Be Paid Before Approval

Purpose:

Prevent Enrollment from being approved until fees are paid.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.

2. Click Validation Rules → New.

3. Rule Name: Fees_Must_Be_Paid_Before_Approval.

4. Error Condition Formula:

```
AND(  
    ISPICKVAL(Enrollment_Status__c, "Approved"),  
    Fees_Paid__c = FALSE  
)
```

5. Error Message: Fees must be paid before approving an enrollment.

6. Error Location: Top of Page (or field as needed).

Click Save.

Setup Home Object Manager

SETUP > OBJECT MANAGER

Enrollment

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Restriction Rules

Scoping Rules

Object Access

Triggers

Flow Triggers

Validation Rules

Rule Name: Fees_Must_Be_Paid_Before_Approval

Active: ☒

Description:

Error Condition Formula

Example: Discount_Percent__c > 0.30 More Examples...

Display an error if Discount is more than 30%

If this formula expression is **true**, display the text defined in the Error Message area

Insert Field Insert Operator

AND(
 ISPICKVAL(Enrollment_Status__c, "Approved"),
 Fees_Paid__c = FALSE
)

Functions

-- All Function Categories --

ABS
ACOS
ADDMONTHS
AND
ASCII
ASIN

Insert Selected Function

ABS(number)
Returns the absolute value of a number, a number without its sign

Help on this function

Check Syntax

Error Message

Example: Discount percent cannot exceed 30%

This message will appear when Error Condition formula is **true**

Error Message: Fees must be paid before approving an enrollment

This error message can either appear at the top of the page or below a specific field on the page

Error Location: ☒ Top of Page ☐ Field

Save Save & New Cancel

Milestone 6 – Approval Process

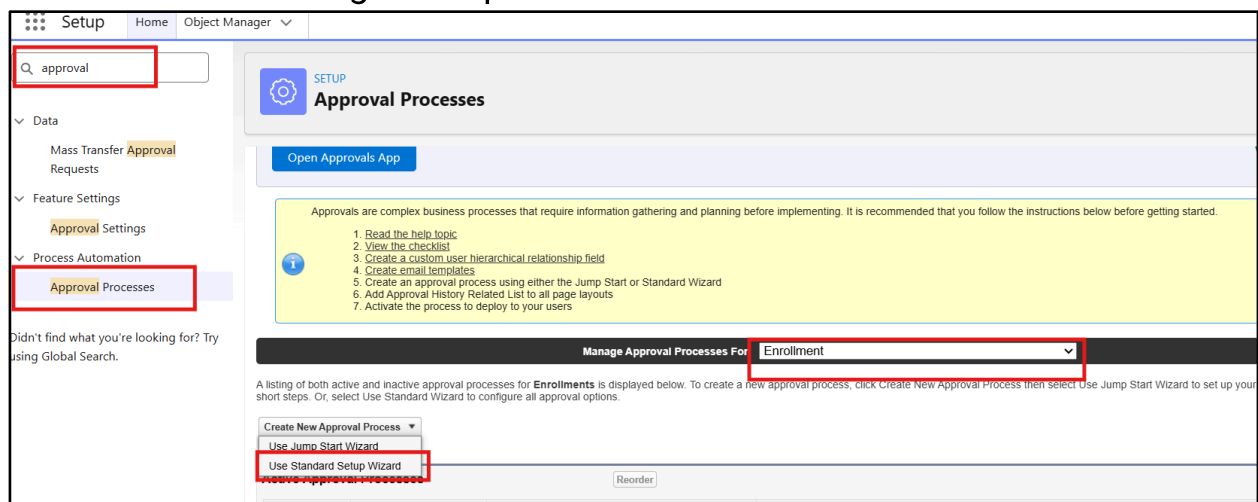
Activity 1: Create Approval Process for Enrollment Requests

Purpose:

Route Enrollment requests to the Training Manager for approval or rejection.

Steps:

1. Go to Setup → Approval Processes → select Enrollment__c.
2. Click Create New Approval Process → Use Standard Setup.
3. Process Name: Approval_Enrollment_Request.
4. Entry Criteria: Enrollment_Status__c = "Requested".
5. Save and configure steps.



Approval Process Edit

Approval Enrollment Request

Step 1. Enter Name and Description

Enter a name and description for your new approval process.

Enter Name and Description

Process Name	Approval Enrollment Request
Unique Name	Approval_Enrollment_Request
Description	Approval process for Enrollment requests — sends to Training Manager

Step 2. Specify Entry Criteria

Previous Save **Next**

If only certain types of records should enter this approval process, enter that criteria below. For example, only expense reports from employees at headquarters should use this approval process.

Specify Entry Criteria

Use this approval process if the following criteria are met:

Field	Operator	Value
Enrollment: Enrollment Status	equals	Requested
--None--	--None--	
--None--	--None--	
--None--	--None--	
--None--	--None--	

AND
AND
AND
AND

6. Add Approval Step:

7. Approver: specific Training Manager user.

SETUP

Approval Processes

Approval Step Edit

Training Manager

Step 1. Enter Name and Description

Save Next Cancel

Enter a name, description, and step number for your new approval step.

Enter Name and Description

Approval Process Name: Approval Enrollment Request

Name	Training Manager
Unique Name	Training_Manager
Description	

Save Next Cancel

8. Under Final Approval Actions:

9. Add Field Update to set Enrollment_Status__c = "Approved".

Field Updates

Define the field update, including the object associated with the workflow rule, approval process, or entitlement process, the field to update, and the value to apply. Note that the fields are shown only for the type that you select.

Field Update Edit Save Save & New Cancel

Identification

Name: Status to Approved

Unique Name: Status_to_Approved

Description:

Object: Enrollment

Field to Update: Enrollment: Enrollment Status

Field Data Type: Picklist

Re-evaluate Workflow Rules after Field Change: ☐

Specify New Field Value

Picklist Options

☐ The value above the current one

☐ The value below the current one

☒ A specific value: Approved

Save Save & New Cancel

10. Under Final Rejection Actions:

11. Add Field Update to set Enrollment_Status__c = "Rejected".

Field Update Edit Save Save & New Cancel

Identification

Name **Set Status to Rejected**

Unique Name **Set_Status_to_Rejected** i

Description

Object Enrollment

Field to Update Enrollment: Enrollment Status

Field Data Type Picklist

Re-evaluate Workflow Rules after Field Change ☐ i

Specify New Field Value

Picklist Options

☐ The value above the current one
☐ The value below the current one
☒ A specific value **Rejected** v

12. Add Email Alert to send rejection email to student.

Email Alert Edit Save Save & New Cancel

Edit Email Alert

Description **Enrollment Rejected**

Unique Name **Enrollment_Rejected** i

Object Enrollment

Email Template **Enrollment Rejected** i

Protected Component ☐

Recipient Type Search: **User** v for: Find

Recipients

Available Recipients

User: Integration User ▲
 User: OrgFarm EPIC
 User: Security User

Selected Recipients

User: sai kumar A ▲

Add ▶
 Remove ◀

You can enter up to five (5) email addresses to be notified.

Additional Emails

geespam198@gmail.com

From Email Address

Current User's email address

☐ Make this address the default From email address for this object's email alerts. [i](#)

Save Save & New Cancel

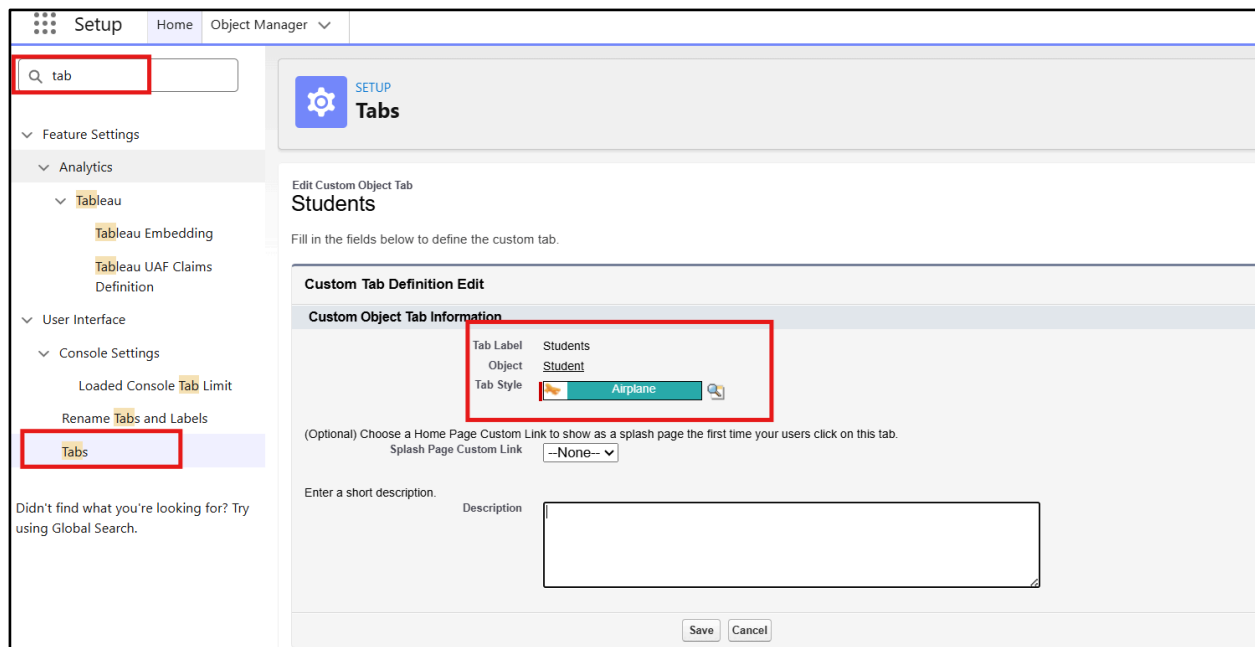
13. Activate the approval process.

Milestone 7- Tabs

Activity 1: Creating a Tab for the Student Object:

Go to Setup.

1. In the Quick Find box, type Tabs.
2. Under Custom Object Tabs, click New.
3. Select Student from the object list.
4. Choose a tab style (icon).
5. Click Next, assign it to all profiles, and then click Save.



Activity 2: Creating Remaining Tabs

Now create the Tabs for the remaining Objects, they are “Courses, Instructors, Enrollment”.

Follow the same steps as mentioned in Activity -1

Milestone 8 - The Lightning App

Activity 1: Creating the Lightning Application for the Project

Navigate to Setup.

Enter App Manager in the Quick Find search box.

Click New Lightning App.

Provide the app details:

App Name: Student Enrollment & Course Management

Developer Name auto-fills

Description: Optional, but can describe the purpose of the app

Click Next and choose the following settings:

App Branding (optional): Add logo or select color theme

Navigation Style: Choose Standard Navigation

Click Next and assign the app to specific profiles (e.g., **Admin, Enrollment Officer, Instructor**).

Click Next to select the items (tabs) to be included in the app. Add:

1. Student
2. Course
3. Instructor
4. Enrollment
5. Reports
6. Dashboards
7. Home
8. Tasks

Click Next and review the app settings.

Finally, click Save & Finish.

The screenshot shows the 'Lightning Experience App Manager' interface. At the top, there's a navigation bar with 'Setup', 'Home', and 'Object Manager'. A search bar contains 'app manager'. Below it, a dropdown menu shows 'Apps' with 'App Manager' selected. On the right, there are buttons for 'New Lightning App' and 'New External Client App'. The main content area is titled 'App Details & Branding' and includes instructions: 'Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.' The page is divided into two columns: 'App Details' and 'App Branding'. In the 'App Details' column, the 'App Name' field is highlighted with a red box and contains 'Student Enrollment & Course Management'. Below it, the 'Developer Name' field contains 'Student_Enrollment_Course_Management'. The 'Description' field is empty. In the 'App Branding' column, the 'Image' field shows a placeholder image. The 'Primary Color Hex Value' field contains '#0070D2'. At the bottom, there's a section for 'Org Theme Options' with a checkbox labeled 'Use the app's image and color instead of the org's custom theme'.

Setup Home Object Manager

Q app manager

SETUP Lightning Experience App Manager

New Lightning App New External Client App

Apps

App Manager

App Details & Branding

Give your Lightning app a name and description. Upload an image and choose the highlight color for its navigation bar.

App Details

* App Name ⓘ

Student Enrollment & Course Management

* Developer Name ⓘ

Student_Enrollment_Course_Management

Description ⓘ

Enter a description...

App Branding

Image ⓘ

Primary Color Hex Value ⓘ

#0070D2

Org Theme Options

☐ Use the app's image and color instead of the org's custom theme

Navigation Items

Choose the items to include in the app, and arrange the order in which they appear. Users can personalize the navigation to add or move items, but users can't remove or rename the navigation items that are available only for phone or only for desktop. These items are dropped from the navigation bar when the app is viewed in a format that the item doesn't support.

Available Items Create ▾

Type to filter list...

Action Hub

- Activation Targets
- Activations
- Agentforce Studio
- All Sites
- Alternative Payment Methods
- Analytics

Selected Items

- Students
- Courses
- Instructors
- Enrollments
- Accounts
- Contacts
- Dashboards

User Profiles

Choose the user profiles that can access this app.

Available Profiles

system Administration

Selected Profiles

- System Administrator
- Standard User

Milestone 9 – Page Layouts and Record:

Activity 1: Create Record Types for Enrollment:

Purpose:

To separate New Enrollment vs Re-Enrollment processes.

Steps:

1. Go to Setup → Object Manager → Enrollment__c.

2. Click Record Types → New.

3. For the first record type:

- Existing Record Type: Master
- Record Type Label: New Enrollment
- Enable for required profiles (Admin, Enrollment Officer).
- Select the default Page Layout → Save & New.

4. For the second record type:

- Record Type Label: Re-Enrollment
- Enable for the same profiles.
- Select the default Page Layout → Save.

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Edit Record Type

Re-Enrollment

Enter a new name for the selected record type and click Save.

Record Type

Record Type Label **Re-Enrollment**

Record Type Name **Re_Enrollment**

Namespace Prefix

Description Used when a student re-enrolls in a course

Active ☒

Save Cancel

SETUP > OBJECT MANAGER

Enrollment

Details

Fields & Relationships

Page Layouts

Lightning Record Pages

Buttons, Links, and Actions

Compact Layouts

Field Sets

Object Limits

Record Types

Related Lookup Filters

Edit Record Type

New Enrollment

Enter a new name for the selected record type and click Save.

Record Type

Record Type Label **New Enrollment**

Record Type Name **New_Enrollment**

Namespace Prefix

Description Used for first-time enrollments

Active ☒

Save Cancel

Activity 2: Separate Page Layout Creators for Each Enrollment Record Type

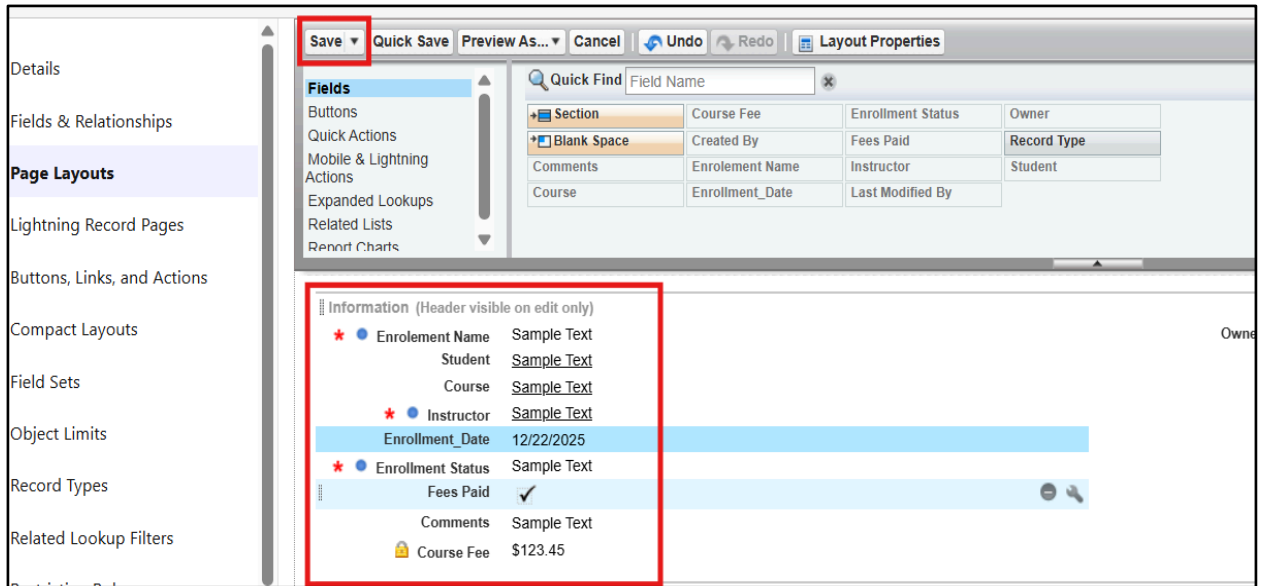
Purpose:

To show different fields for New Enrollment vs Re-Enrollment.

Steps:

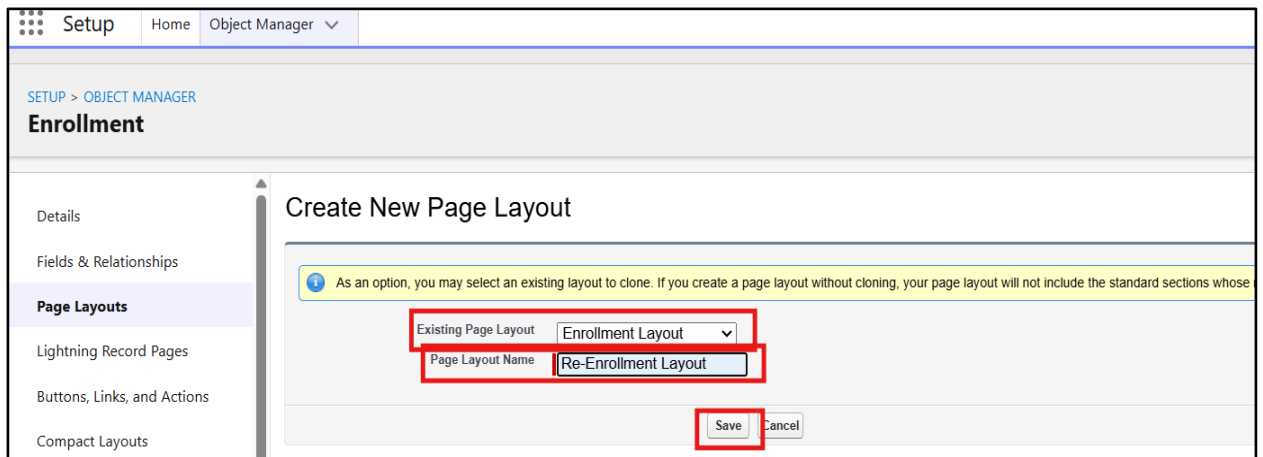
- 1.Enrollment__c object → click Page Layouts.
- 2.Click New to create a layout for New Enrollment:
 - Clone from existing Enrollment Layout → name it New Enrollment Layout → Save.
 - Keep fields relevant for first-time enrollment: Student, Course, Instructor, Enrollment Date, Status, Fees, etc.
 - Remove unnecessary re-enrollment fields if any.
 - Click Save.

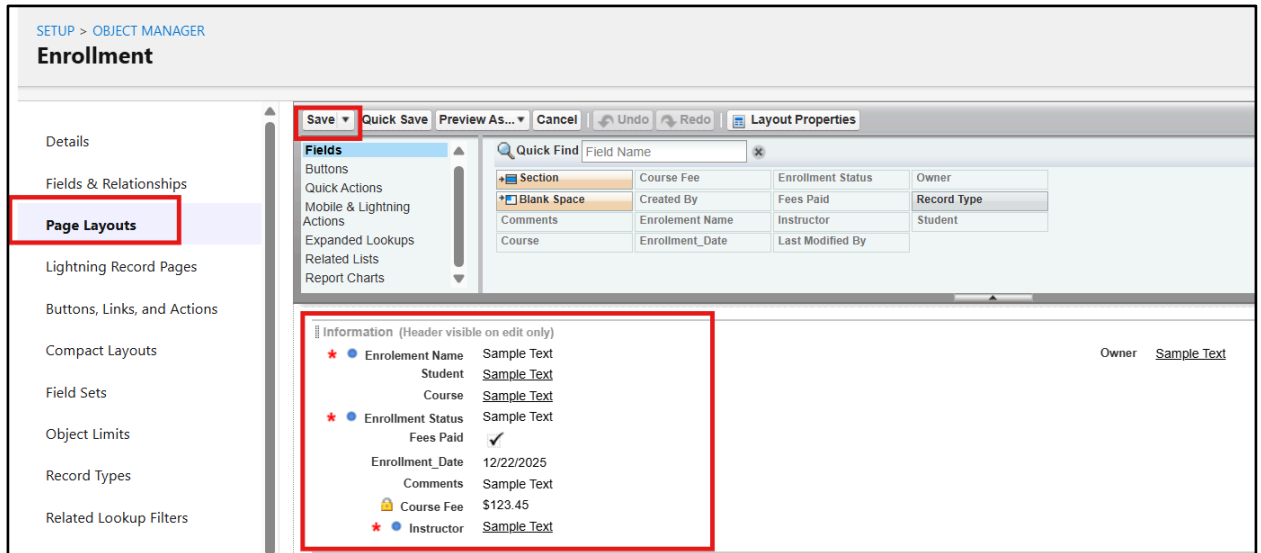
The screenshot shows the Salesforce 'Create New Page Layout' interface for the 'Enrollment' object. On the left sidebar, the 'Page Layouts' option is highlighted with a red box. The main area is titled 'Create New Page Layout'. A yellow information banner at the top states: 'As an option, you may select an existing layout to clone. If you create a page layout without cloning, your page layout will not include the standard fields.' Below this, the 'Existing Page Layout' dropdown menu is set to 'Enrollment Layout' and is highlighted with a red box. The 'Page Layout Name' text input field contains 'New Enrollment Layout' and is also highlighted with a red box. At the bottom right, the 'Save' button is highlighted with a red box, next to a 'Cancel' button.



3. Again, click New to create Re-Enrollment Layout:

- Clone existing layout → name it Re-Enrollment Layout → Save.
- Include fields like previous Enrollment reference (if any), Re-Enrollment Reason, Status, Fees, etc.
- Click Save.





4. Click Page Layout Assignment → Edit Assignment.

5. For each profile and record type:

- New Enrollment → use New Enrollment Layout.

6. Re-Enrollment → use Re-Enrollment Layout.

Click Save.

Activity 3: Enrollment Related Lists to Parent Objects:

Purpose:

To see all Enrollments from Student, Course, and Instructor records.

a) Student → Enrollment Related List

Go to Setup → Object Manager → Student__c.

Click Page Layouts → open the main layout.

In the palette, select Related Lists.
Drag Enrollments related list onto the layout.
Customize columns if needed → Save.

b) Course → Enrollment Related List

Go to Object Manager → Course__c → Page Layouts.

Open layout → add Enrollments related list.

Save.

c) Instructor → Enrollment Related List

Go to Object Manager → Instructor__c → Page Layouts.

Open layout → add Enrollments related list.

Save.

Milestone 10 – Automation with Flow

Activity 1: Auto-Populate Enrollment Date (Record-Triggered Flow)

Purpose:

Set Enrollment_Date__c to today when Enrollment is created.

Steps:

1. Go to Setup → Flows → New Flow.
2. Select Record-Triggered Flow → Create.
3. Object: Enrollment__c.
4. Trigger: A record is created.
5. Optimize for Actions and Related Records → Done.

Select Object

Select the object whose records trigger the flow when they're created, updated, or deleted.

* Object

Enrollment

Configure Trigger

Trigger the Flow When:

☒ A record is created

☐ A record is updated

☐ A record is created or updated

☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of records that trigger the flow to conserve your org's resources.

Condition Requirements

None ▼

6. Add Update Records element:
 - Label: Set Enrollment Date.
 - Use a record that triggered the flow.
 - Field: Enrollment_Date__c → Value: \$Flow.CurrentDate.

7. Save as SetEnrollmentDate → Activate.

 Update Records

* Label

Set Enrollment Date

* API Name ⓘ

Set_Enrollment_Date

Description

* How to Find Records to Update and Set Their Values

☒ Use the enrollment record that triggered the flow
 ☐ Update records related to the enrollment record that triggered the flow
 ☐ Use the IDs and all field values from a record or record collection
 ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record

None—Always Update Record ▼

Set Field Values for the Enrollment Record

Field

📅

Enrollment_Date

×

←

Value

📅

Running Flow Interview > CurrentDate

×

🗑️

Activity 2: Send Email When Enrollment Approved

Purpose:

When Enrollment_Status__c = Approved, send email, alert admin, and set Student to Active.

Steps (high level for document):

1. Create Email Template and Email Alert for student notification.
2. Go to Flows → New → Record-Triggered Flow.
3. Object: Enrollment__c, Trigger on update.
4. Condition: Enrollment_Status__c Equals "Approved".

The screenshot shows the 'Configure Start' section of a Salesforce Flow. At the top, there is a 'Configure Start' button with a play icon. Below it, the 'Object' is set to 'Enrollment'. The 'Configure Trigger' section is active, showing four options for 'Trigger the Flow When:'. The selected option is 'A record is created or updated'. Below this, the 'Set Entry Conditions' section is visible. It includes a description about minimizing unnecessary flow executions and a recommendation to define entry conditions. A 'Condition Requirements' dropdown is set to 'All Conditions Are Met (AND)'. At the bottom, there is a table for defining conditions with columns for 'Field', 'Operator', and 'Value'. The first condition is defined with 'Enrollment Status' as the field, 'Equals' as the operator, and 'Approved' as the value.

Configure Start

Enrollment

Configure Trigger

Trigger the Flow When:

- ☐ A record is created
- ☐ A record is updated
- ☒ A record is created or updated
- ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements

All Conditions Are Met (AND)

Field	Operator	Value
Enrollment Status	Equals	Approved

5. Add Get Records to fetch related Student__c.

 Get Records ✕

* Label

Get Student

* API Name ⓘ

Get_Student

Description

Get Records of This Object

* Data Source

☐ Data Cloud Object

☒ Salesforce Object

* Object

Student

Filter Student Records

Condition Requirements

All Conditions Are Met (AND) ▼

Field

Operator

Value

Status ✕

Equals ▼

Aa Active ✕

✕

+ Add Condition

6. Add Action (Email Alert) to send email to student.

 Enrollment Approved EmailAlert ×

* Label

Send Student Email

* API Name 

Send_Student_Email

Description

Use values from earlier in the flow to set the inputs for the "Enrollment Approved EmailAlert" email alert. To use its outputs later in the flow, store them in variables.

Set Input Values

 * Record ID 

 Triggering Enrolement__c > Record ID ×

7. Add Action to notify Admin if needed.
8. Add Update Records to set Student__c.Status__c = "Active".


Update Records
✕

* Label

Set Student Status Active

* API Name ⓘ

Set_Student_Status_Active

Description

* How to Find Records to Update and Set Their Values

☒ Use the enrollment record that triggered the flow
 ☐ Update records related to the enrollment record that triggered the flow
 ☐ Use the IDs and all field values from a record or record collection
 ☐ Specify conditions to identify records, and set fields individually

Set Filter Conditions

Condition Requirements to Update Record

None—Always Update Record ▼

Set Field Values for the Enrollment Record

Field

Value

Aa Comments ✕

←  True ✕



9. Save as Enrollment Approved Action → Activate.

Activity 3: Auto-Assign Instructor Based on Course Category

Purpose:

Assign Instructor automatically based on Course category.

Rules:

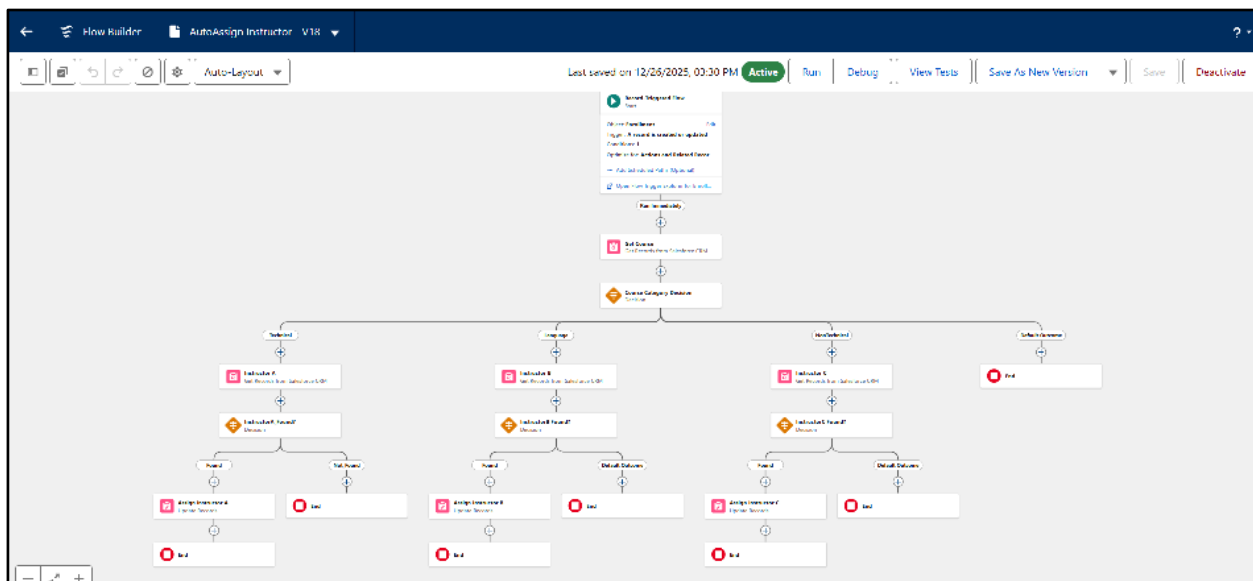
Technical → Instructor A

Language → Instructor B

Non-Technical → Instructor C

Steps:

1. Prepare Instructor__c records with codes (e.g., INSTR_A, INSTR_B, INSTR_C).
2. Create Record-Triggered Flow on Enrollment__c (created/updated).
3. Add Get Records to fetch related Course__c.
4. Add Decision element on Course__r.Category__c:
5. Technical / Language / Non-Technical outcomes.
6. For each outcome:
7. Get Instructor record (A/B/C) based on code or name.
8. Update Records: set Instructor__c = Get_Instructor_X.Id.
9. Save as AutoAssign_Instructor → Activate.



Activity 4: Auto-Create Task for Follow-Up

Purpose:

When Enrollment is "Requested", create a follow-up task for Admin.

Steps:

1. Create Record-Triggered Flow on Enrollment__c.
2. Trigger: Record created or updated; condition Enrollment_Status__c = "Requested".

The screenshot shows the 'Configure Start' dialog for a new flow. The 'Object' field is set to 'Enrollment'. Under 'Configure Trigger', the option 'A record is created or updated' is selected. The 'Set Entry Conditions' section shows 'All Conditions Are Met (AND)' as the condition requirements. A single condition is defined: 'Enrollment Status' equals 'Requested'.

Configure Start

* Object
Enrollment

Configure Trigger

Trigger the Flow When:

- ☐ A record is created
- ☐ A record is updated
- ☒ A record is created or updated
- ☐ A record is deleted

Set Entry Conditions

Specify entry conditions to reduce the number of records that trigger the flow and the number of times the flow is executed. Minimizing unnecessary flow executions helps to conserve your org's resources.

If you create a flow that's triggered when a record is updated, we recommend first defining entry conditions. Then select the **Only when a record is updated to meet the condition requirements** option for When to Run the Flow for Updated Records.

Condition Requirements
All Conditions Are Met (AND)

Field	Operator	Value
Enrollment Status	Equals	Requested

3. Add Get Records to fetch Admin user (by Username / Profile).
4. Add Formula (Date): TODAY() + 2 for due date.


Get Records
×

Description

Get Records of This Object

*** Data Source**

☐ Data Cloud Object
 ☒ Salesforce Object

*** Object**

Filter User Records

Condition Requirements

All Conditions Are Met (AND)
 ▼

Field	Operator	Value
Aa Username ×	Equals ▼	geespam198430@agentforce.com 🔍 🗑️

+ Add Condition

5. Add Create Records for Task:
6. Subject: Follow up on enrollment request.
7. OwnerId: Admin user Id.
8. WhatId: Enrollment Id.
9. ActivityDate: formula Today + 2.

 Create Records

Create a Record of This Object

* Object
 Task

Set Field Values for the Task

Field	Value
Due Date Only	DueDateTodayPlus2
Description	Auto-created follow-up for enrollment request
Priority	Normal
Status	Not Started
Subject	Follow up on enrollment request
Related To ID	Triggering Enrolement_c > Record ID

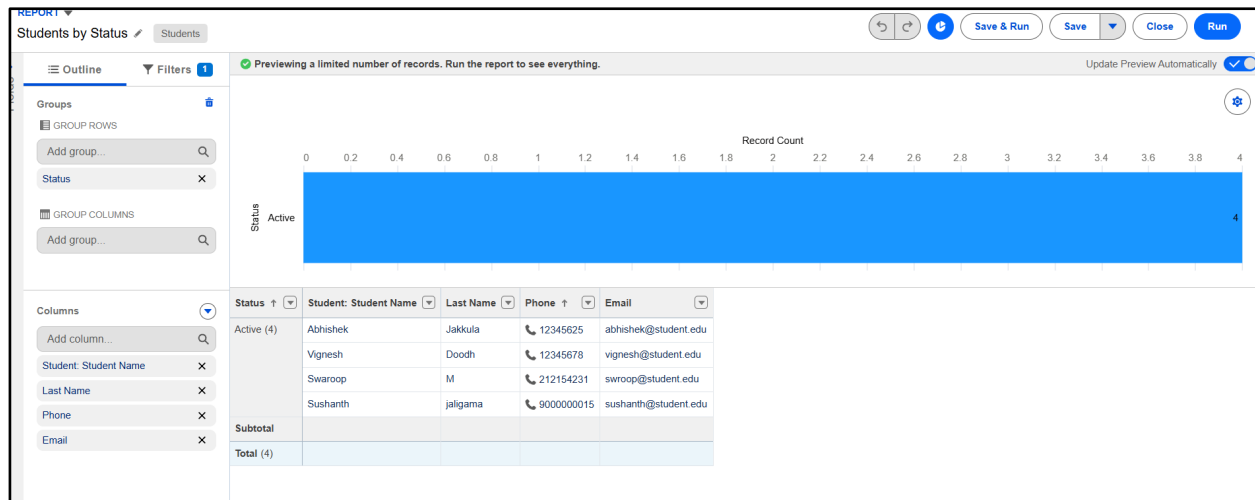
10. Save as Create_Followup_Task → Activate.

Milestone 11 – Reports:

Activity 1: Create “Students by Status” Report

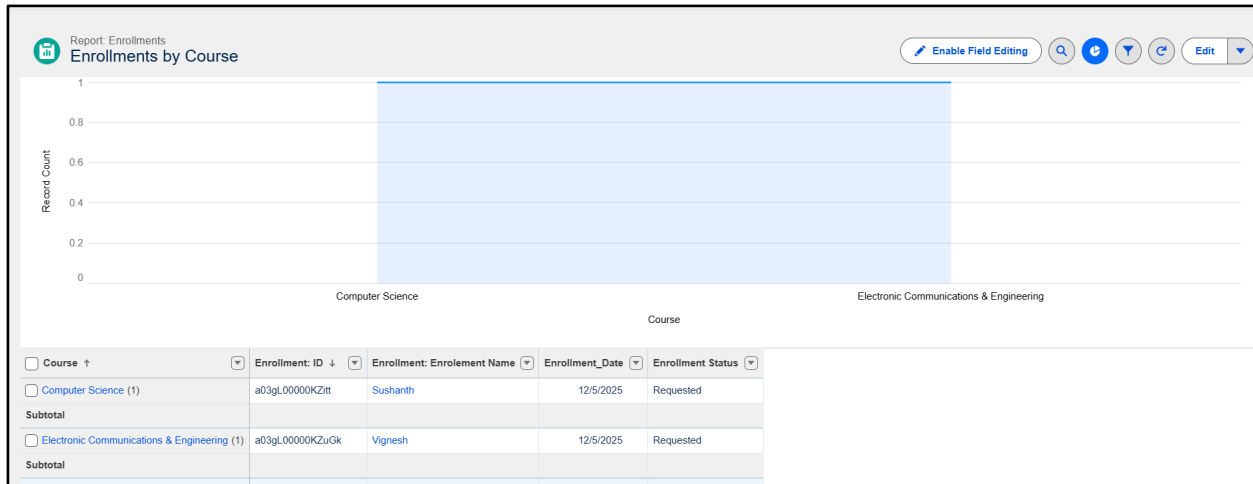
- 1.Go to Reports → New Report.
- 2.Report Type: Students.
- 3.Group rows by Status__c.
- 4.Add columns as needed (Name, Email, Phone).

5. Save as **Students by Status**.



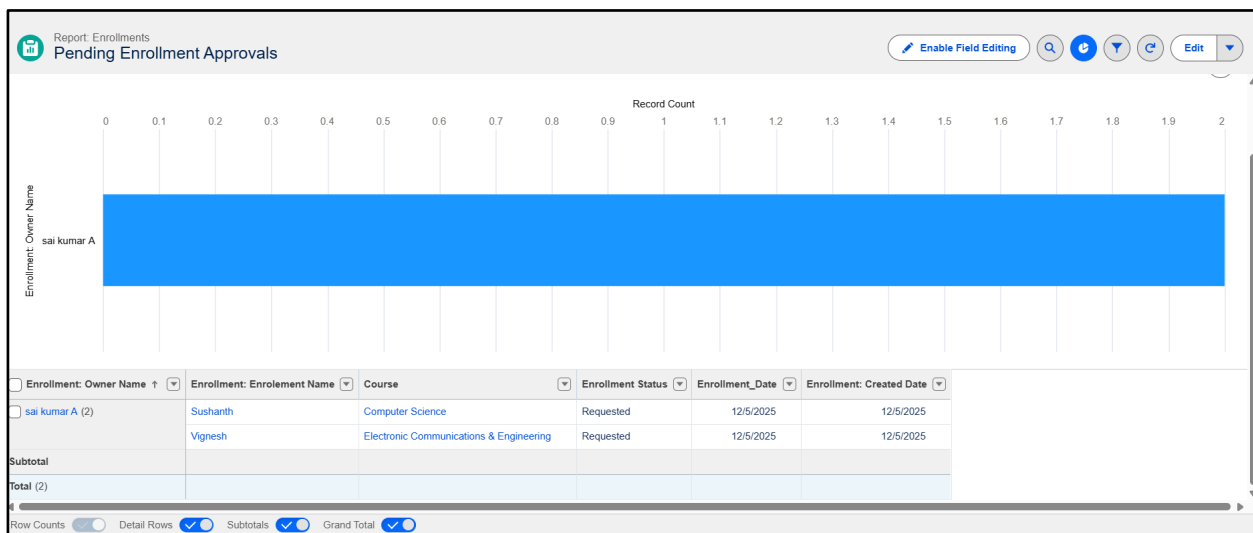
Activity 2: Create “Enrollments by Course” Report

1. New Report → Report Type: **Enrollments**.
2. Grouprows by **Course__r.Name**.
3. Show Record Count per course.
4. Save as **Enrollments by Course**



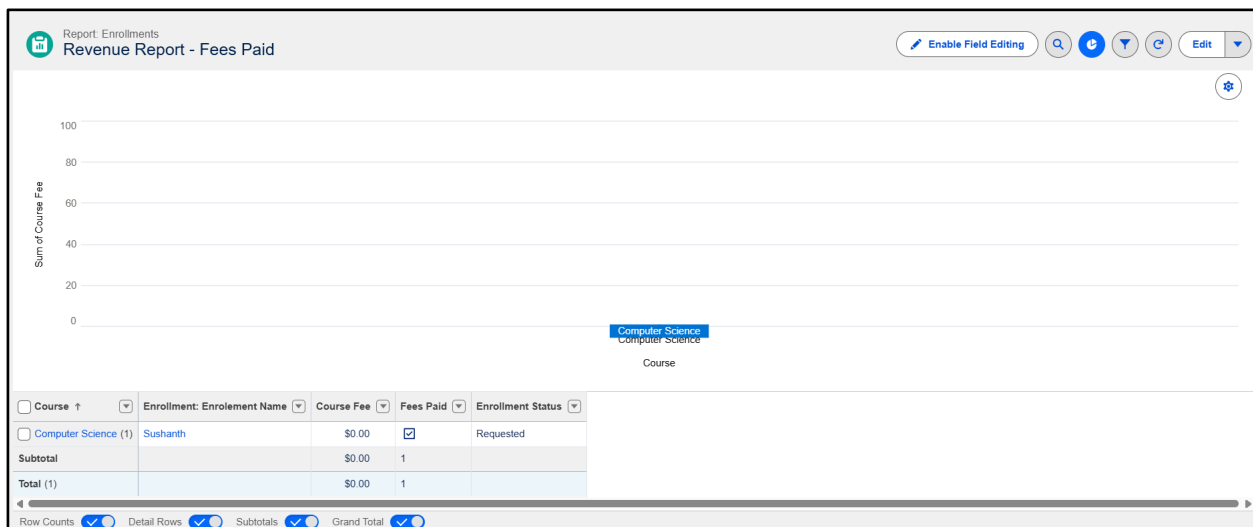
Activity 3: Create "Pending Enrollment Approvals" Report

1. New Report → **Enrollments** report type.
2. Filter: **Enrollment_Status__c = "Requested"**.
3. Add columns (Student, Course, Owner, Created Date).
4. Save as **Pending Enrollment Approvals**.



Activity 4: Create “Revenue Report (Fees Paid)” Report

- 1.New Report → **Enrollments** report type.
- 2.Filter: **Fees_Paid__c = TRUE**.
- 3.Group by **Course__r.Name**.
- 4.Summarize **Course_Fee__c** or similar fee field.
- 5.Save as **Revenue Report – Fees Paid**.

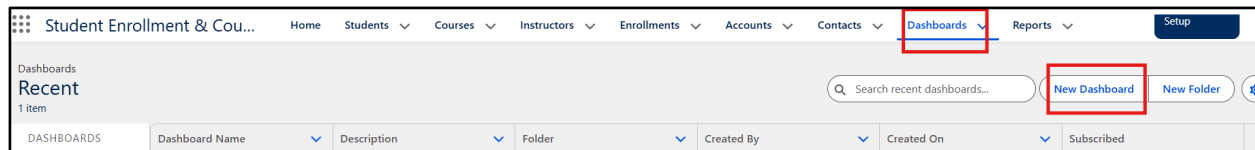


Milestone 12 – Dashboards:

Activity 1: Create “Student Managemet Dnashboard”

Steps:

1. Go to **Dashboards** → **New Dashboard**.
2. Name: **Student Management Dashboard**.



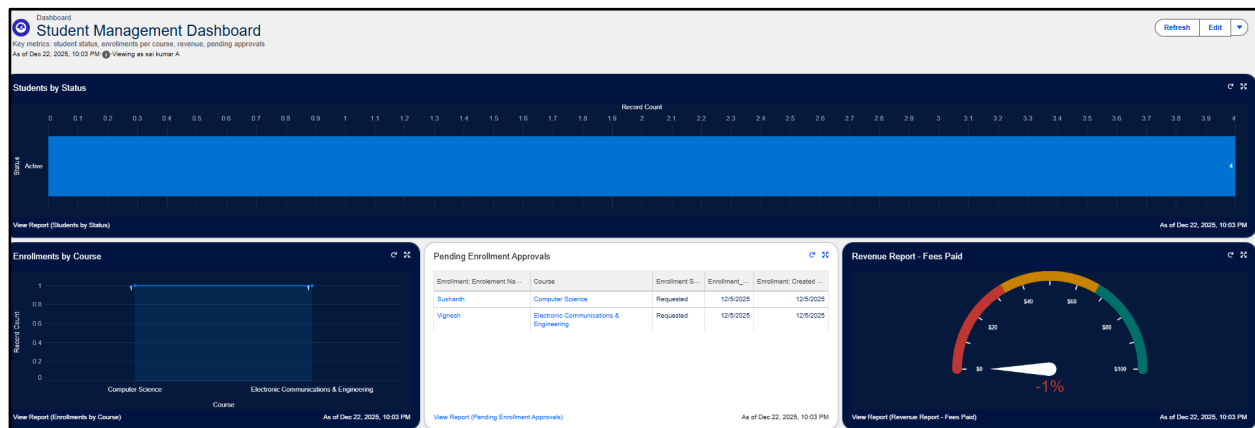
3. Select folder → **Create**.

4. Select the new Dashboard folder and Save.

Add components:

- **Donut Chart** → using **Students by Status** report.
- **Bar Chart** → using **Enrollments by Course** report.
- **Gauge** → using **Revenue Report – Fees Paid** (sum of revenue field).
- **Table** → using **Pending Enrollment Approvals** report.

4. Arrange components on canvas.



Click **Save** → **Done**.

Milestone 13 – Security Setup

Activity 1: Configure Profiles

):

1. Go to **Setup** → **Profiles** → **New** (or Clone)**Profiles Required:**
 - **Admin** – Full access.
 - **Enrollment Officer** – Create/Update Enrollments.
 - **Instructor** – Read-only access to assigned Enrollments.
2. **Steps** (example for Enrollment Officer fName: **Enrollment Officer**).
3. In **Object Settings**:
 - **Enrollment__c**: Read, Create, Edit.
 - **Student__c** / **Course__c**: Read (and Edit if required).
4. Save.

Repeat similarly for **Instructor** profile with **Read-Only** on Enrollments

Activity 2: Create Permission Sets

1. Go to **Setup** → **Permission Sets** → **New**.
2. Create **Custom Object Access** (additional access to project objects if needed).
3. Create **Dashboard Editor** (Create/Customize Reports & Dashboards permissions).
4. Assign to relevant users.

Activity 3: Configure Sharing & OWD

Steps:

1. Go to **Setup** → **Sharing Settings**.
2. Set **Organization-Wide Defaults**:
 - **Students (Student__c)**: Public Read/Write.
 - **Courses (Course__c)**: Public Read/Write.
 - **Enrollments (Enrollment__c)**: Private.
3. Save.

For Instructor access to assigned Enrollments, optionally:

- Add a **User lookup** on Enrollment (Instructor User).
- Use a simple **Record-Triggered Flow** to set **OwnerId = Instructor User** so they can see only their own records.

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce user interface to verify the proper functioning of all implemented features of the Student Enrollment & Course Management System (SECMS). These include student, course, instructor, and enrollment record creation; validation rule checks for mandatory fields, eligibility, and enrollment constraints; verification of record type behavior for different course categories; flow execution for automated enrollment approvals and seat availability checks; and Apex trigger functionality to ensure automatic confirmation and record updates. Additionally, screenshots of generated reports and dashboards were documented to validate accurate data visualization, enrollment tracking, and performance monitoring. Submit the Screenshots while testing.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Centralized data
- Automated workflows
- Real-time reporting

Disadvantages

- Salesforce dependency
- Requires initial configuration effort

9. Conclusion:

The Salesforce Student Enrollment & Course Management System successfully streamlines the entire student academic lifecycle—from student registration and course creation to enrollment management, instructor assignment, approvals, revenue tracking, and reporting. By leveraging Salesforce’s automation tools such as validation rules, flows, approval processes, and dashboards, the system improves data accuracy, operational efficiency, and decision-making. This project demonstrates how Salesforce can transform manual academic processes into a scalable, reliable, and user-friendly digital solution.

10. FUTURE SCOPE

- Payment gateway integration
- Student self-service portal
- Analytics-driven insights

- Mobile access support

11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

Thank You

